

Rafat Ashraf Joy

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SUMMARY

Fourth-year Computer Science PhD student with research expertise in causal inference and causal discovery, time-series modeling, and deep generative learning. Developing statistical and machine learning methods for causal reasoning in high dimensional and incomplete data. Seeking a PhD research internship in machine learning, causal inference, or applied research.

RESEARCH INTERESTS

Causal Inference; Deep Generative Models; Graph Neural Networks, Time Series Data.

EDUCATION

University of Illinois Chicago

PhD, Computer Science

Aug 2023 – May 2028 (expected)

Chicago, IL, USA

- GPA: 4.00 / 4.00
- Relevant Coursework: Machine Learning on Graphs, Causal Inference, Energy Efficient Deep Learning, Algorithmic Fairness and Responsible AI.

RESEARCH EXPERIENCE

Graduate Research Assistant

University of Illinois Chicago (Advisor: Elena Zheleva)

May 2025 – Present

Chicago, IL, USA

- Developed TriOpt, a scalable causal discovery algorithm that exploits causal ordering and Sherman–Morrison rank-one updates to reduce computational cost for high-dimensional structure learning. Achieved up to 42× speedup over NOTEARS, 25× over GOLEM, and 16× over DAGMA on 1,000-variable benchmarks.
- Developed SCORE-EM, an EM-based causal discovery framework for missing-at-random data using MALA-based conditional imputation and score-matching-based ordering; outperforms MissScore, MissDAG, and MissOTM across synthetic benchmarks.
- Applied causal discovery methods including NOTEARS, DAGMA, PC, FCI, and GES to oral microbiome and psychiatric digital-phenotyping data in interdisciplinary collaborations with UIC Dentistry and Bioengineering.

Graduate Research Assistant

University of Illinois Chicago (Advisor: Pedram Rooshenas)

Aug 2023 – Dec 2024

Chicago, IL, USA

- Investigated diffusion and energy-based generative methods for improving optimization and convergence of Physics-Informed Neural Networks for solving partial differential equations.

PROFESSIONAL EXPERIENCE

Junior Software Engineer

Dynamic Solution Innovators

Apr 2023 – Jul 2023

Dhaka, Bangladesh

- Contributed to backend development of an internet service provider's subscription platform using the Ruby on Rails framework.
- Implemented automated invoice PDF generation with the Prawn and Receipts gems.

TECHNICAL SKILLS

- **Languages:** Python, C++, Java, SQL
- **ML/AI:** PyTorch, PyTorch Geometric, Hugging Face, scikit-learn, NumPy, SciPy, CuPy
- **Causal ML:** DoWhy, EconML, causal-learn, gCastle, CausalNex
- **GenAI/LLM:** Retrieval-Augmented Generation (RAG), vector databases, embeddings, LLM quantization
- **Systems & Compute:** Linux, Git, Docker, SLURM/HPC, GCP, distributed training (DDP/FSDP)

- **Methods:** Causal inference and discovery, diffusion models, VAEs, GANs, GNNs, time-series modeling

PUBLICATIONS

- **Joy, Rafat Ashraf** and Elena Zheleva. “TICONV: Accurate Temporal Causal Discovery at Scale” Under Review at AAAI 2027.
- **Joy, Rafat Ashraf**, Huiling Liao and Elena Zheleva. “SCORE-EM: Ordering Based Causal Discovery with Missing-at-Random Data” Under Review at ICLR 2027.
- **Joy, Rafat Ashraf** and Elena Zheleva. “TriOpt: A Scalable Algorithm for Linear Causal Discovery.” Under Review at AAAI 2027.
- **Joy, Rafat Ashraf**. “Fine Tuning the Prediction of the Compressive Strength of Concrete: A Bayesian Optimization Based Approach.” *2021 International Conference on Innovations in Intelligent Systems and Applications (INISTA)*, IEEE, 2021.
- **Joy, Rafat Ashraf**. “An Interpretable CatBoost Model to Predict the Power of Combined Cycle Power Plants.” *2021 International Conference on Information Technology (ICIT)*, IEEE, 2021.

PROJECTS

- **Graph-NERD**. Developed a GNN-based relational causal discovery framework integrating FastGAE structural pruning to reduce the search space and improve scalability of structural equation modeling on relational datasets.
- **NuQuant**. Calibration-free, neuron-level quantization method for large language models (LLMs) built on a Mixture-of-Experts (MoE) transformer architecture; evaluated on Mixtral 8x7B, reducing model footprint for efficient foundation-model inference and deployment with minimal degradation in downstream task performance.
- **Keshik**. Published pip package addressing class imbalance in tabular data via a Denoising Diffusion Probabilistic Model (DDPM). Implemented in PyTorch with oversampling APIs applicable to fraud detection, anomaly detection, and churn prediction.
- **Super-Resolution GAN for Precipitation Downscaling**. Designed a super-resolution Generative Adversarial Network to upscale low-resolution (16x16) South Asian precipitation fields to high resolution (64x64), improving the fidelity of meteorological data for downstream forecasting and analysis.
- **Temporal Causal Graph Discovery in HPC Network Traffic Simulations**. Identified causal drivers of variation in multivariate temporal HPC network traffic data from Argonne National Laboratory to improve surrogate model forecasting. Employed Neural Additive Vector Autoregression (NAVAR) to capture nonlinear causal influence and evaluated CausalFormer, combining multi-kernel causal convolution with attention layers, using Regression Relevance Propagation (RRP) to construct the temporal causal graph.

TEACHING EXPERIENCE

Graduate Teaching Assistant

University of Illinois Chicago

Jan 2024 – April 2026

Chicago, IL, USA

- Served as teaching assistant for CS 480 (Database Systems), CS 251 (Data Structures) and CS 418 (Data Science).